

No negative Waring singular series

Disproof of Conjecture 00000004152

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Abstract

Every finite real Waring singular series defined by the Euler product of local representation densities is nonnegative. Each local factor is nonnegative as well. Thus fourth powers cannot have a negative singular series or a negative local factor, contrary to both assertions of the conjecture.

Statement and conventions

The source defines the Waring singular series as the local Euler product for representations as sums of k th powers. It asserts that the smallest k admitting a negative value is 4, and that a negative value for fourth powers comes from the factor at a single prime.

Fix positive integers k, s and a target integer $a \geq 0$. For $q \geq 1$, put

$$R_{k,s}(a; q) = \# \left\{ (x_1, \dots, x_s) \in (\mathbb{Z}/q\mathbb{Z})^s : \sum_{i=1}^s x_i^k = a \pmod{q} \right\}.$$

Whenever it exists as a finite real number, the standard local density is

$$\sigma_p = \lim_{h \rightarrow \infty} p^{h(1-s)} R_{k,s}(a; p^h).$$

If these local factors exist and the prime-ordered product converges, the Euler-product singular series is

$$\mathfrak{S}_{k,s}(a) = \lim_{N \rightarrow \infty} \prod_{\substack{p \leq N \\ p \text{ prime}}} \sigma_p.$$

These are representation counts, local-density limits, and ordinary Euler-product limits. We make no assertion about a regularized value assigned to a divergent expression; the source specifies no such regularization.

Theorem 1. *Every existing finite local factor σ_p is nonnegative. Every existing finite Euler-product singular series $\mathfrak{S}_{k,s}(a)$ is nonnegative.*

Proof. The cardinality $R_{k,s}(a; p^h)$ is a nonnegative integer, and $p^{h(s-1)} > 0$. Hence every normalized count

$$\frac{R_{k,s}(a; p^h)}{p^{h(s-1)}}$$

is nonnegative. The closedness of $[0, \infty)$ gives $\sigma_p \geq 0$ whenever the limit exists. Every finite product of such factors is therefore nonnegative. Applying the same closedness argument to the sequence of finite products proves $\mathfrak{S}_{k,s}(a) \geq 0$. $\square$

Consequence

The theorem applies to every k , including $k = 4$. Thus no fourth-power instance has a negative Euler-product singular series, and no prime supplies a negative local factor. A zero factor, caused by a local obstruction, can give a zero product; it cannot change the sign. Failure of convergence does not create a negative value either. Consequently there is no smallest power with the property alleged in the source, and the conjecture is false.

Formal verification

The Lean project defines `WaringSolutions` as the actual finite subtype of residue tuples satisfying the congruence. The tuple entries are `Fin q`, the canonical representatives $0, \dots, q - 1$; `solutionCount` is their `Nat.card`. The density uses $h = e + 1$, which discards only the exponent-zero term.

`IsLocalFactor` uses `Filter.Tendsto` of those normalized counts. No sign condition is included in this definition. `eulerPartial` is the product over the actual primes in $\{0, \dots, N\}$. `IsWaringSingularSeries` requires the existence of the local limits and the limit of these products. The theorem `waring_singular_series_nonnegative` proves the sign from these definitions, and the two final theorems negate the negative fourth-power global and local assertions separately.

The project is pinned to Lean 4.19.0 and a fixed Mathlib commit. Its proof contains no added axioms or proof placeholders; the printed theorem dependencies are the standard `propext`, `Classical.choice`, and `Quot.sound`. The result is conditional only on the existence of the limits that define the alleged negative value. Establishing convergence for all possible parameter choices is unnecessary for this disproof.

Source

The Last Math Competition, Conjecture 00000004152. The submission includes the exact bilingual source and its SHA-256 provenance record.